

Benchmark Report — California 2021 Greedy Kernel

Date: 2026-03-13

Dataset: California 2021

Risk map: Pyrologix ignition probability (static, trained on 2006–2020, no data leakage for 2021)

Overview

This report summarizes a placement-only benchmark on the California 2021 dataset using the new **StationMax greedy-kernel** sensor-placement strategy.

Main ideas:

- Coverage from **different charging stations does not add** at a cell.
- For each cell, only the **best contributing station** is counted.
- Same-station multi-drone coverage is modeled through **precomputed heuristic coordinated kernels** for $k = 1..7$ drones.
- Candidate station-cell pairs are pruned to those within **one-way reach** $\text{floor}(\text{max_battery_time} / 2) = 3$ operational cells.

This report covers the three budget levels requested:

- **20M**
- **100M**
- **500M**

The 20M and 100M runs produced meaningful placements. The 500M run did not find a useful incumbent within the 10-minute cap and therefore returned an empty placement; this is documented explicitly below.

Heuristic Kernel Construction

For each candidate charging station:

1. Build the local 7×7 operational patch corresponding to one-way reach ≤ 3 .
2. Enumerate masked shortest paths from the station to every reachable cell in that patch.
3. Greedily pick up to $K_{\text{max}} = 7$ paths:
 - drone 1 takes the path that covers the most still-uncovered cells,
 - each additional drone takes the path with the largest remaining marginal gain.
4. The k -drone kernel is the **union** of the first k selected paths.

This is an optimistic coordinated-routing surrogate: drones from the same station are encouraged to spread out, but the preprocessing stays cheap because it only solves tiny local shortest-path problems.

Across stations, the optimizer uses a **max-over-stations** coverage model instead of linear addition.

Common Setup

Parameter	Value
Data grid	1309 × 805 (~1 km/cell)
Operational grid	261 × 161 (5 km/cell)
Drone speed	600 m/min
Coverage radius	2900 m
Battery	1 h = 7 operational substeps
One-way reach used in pruning	3 operational cells
Candidate pre-filtering	top 20% by risk / coverage potential
Solver	Gurobi
Time limit per run	600 s default; 1800 s for the final 100M rerun

Fire Locations

The same California 2021 fire sample/background map is used for all three budgets. Background is the Pyrologix ignition probability map.

Summary Table

Budget	Status	Gap	Preprocess	Model creation	Solve	Allocation summary
20M	TIME_LIMIT	0.37%	2.37 s	3.59 s	601.61 s	0 sensors, 40 stations, 280 drones
100M	TIME_LIMIT	13.0%	2.61 s	6.59 s	1801.69 s	0 sensors, 214 stations, 1358 drones
500M	TIME_LIMIT	infeasible	3.02 s	3.80 s	602.20 s	no useful incumbent found

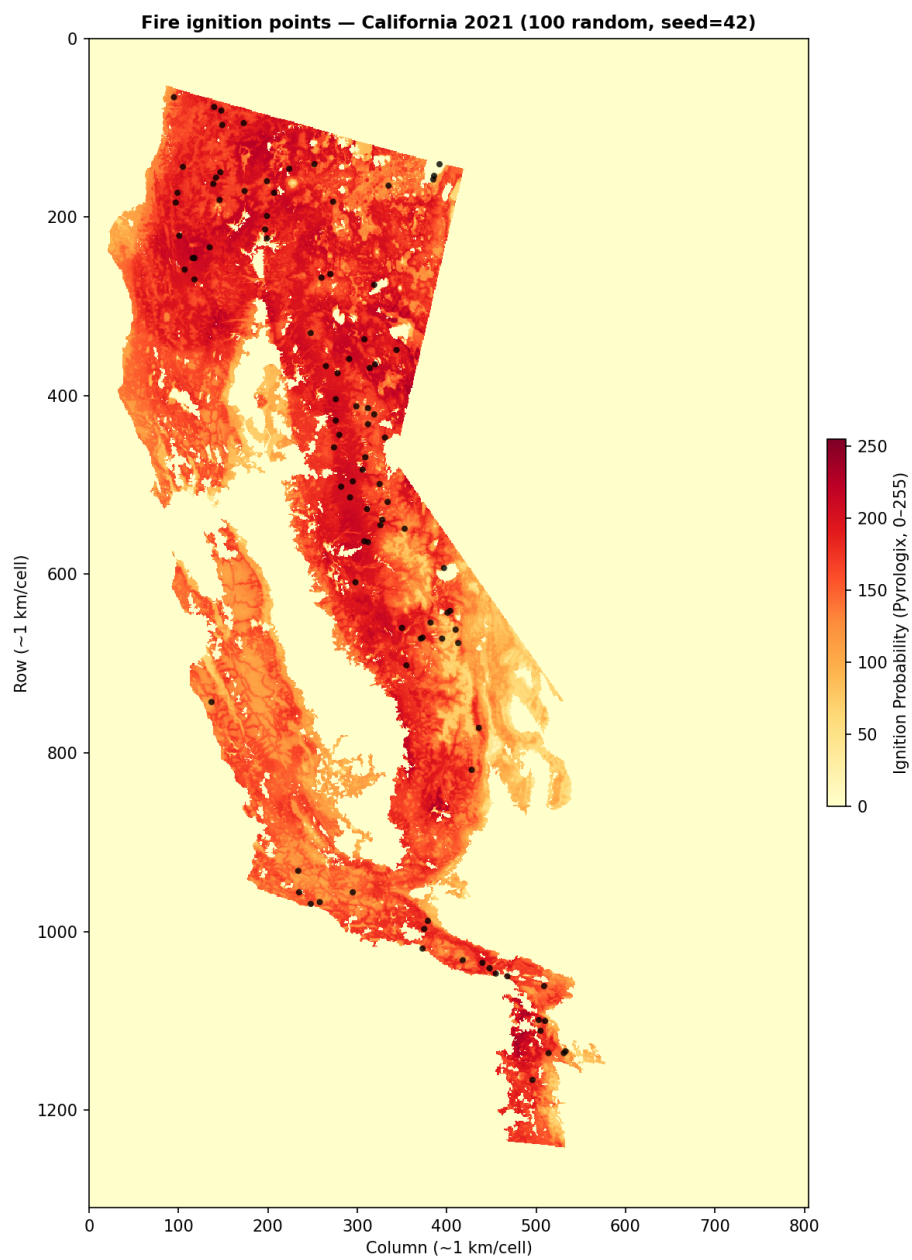


Figure 1: California 2021 fire locations

Notes:

- The 500M run returned objective -0.0 and an empty placement. That should be treated as **failure to find a meaningful feasible solution within the cap**, not as a valid benchmark result.
- The 100M entry above reflects the **fixed 30-minute rerun** (1800 s cap) with two changes relative to earlier runs: masked cells no longer contribute to the objective, and the candidate filter keeps the top **50%** of cells instead of the top 20%.

20M Budget

Allocation

Component	Count	Unit cost	Subtotal
Ground sensors	0	100k	0.00M
Charging stations	40	150k	6.00M
Drones	280	50k	14.00M
Total			20.00M

Every selected charging station received the full cap of **7 drones**.

Solver stats

Metric	Value
Feasible station-cell pairs	49,194
Station-level drone binaries	15,666
Preprocessing	2.37 s
Model creation	3.59 s
Solving	601.61 s
Termination	TIME_LIMIT
MIP gap	0.37%

Plots

100M Budget

Allocation

**Pyrologix + cluster zones + fires [GaussianBudget20M_StationMax_261x161_mean]
8 clusters · 931 fires (246 discoverable, 685 non-disc)**

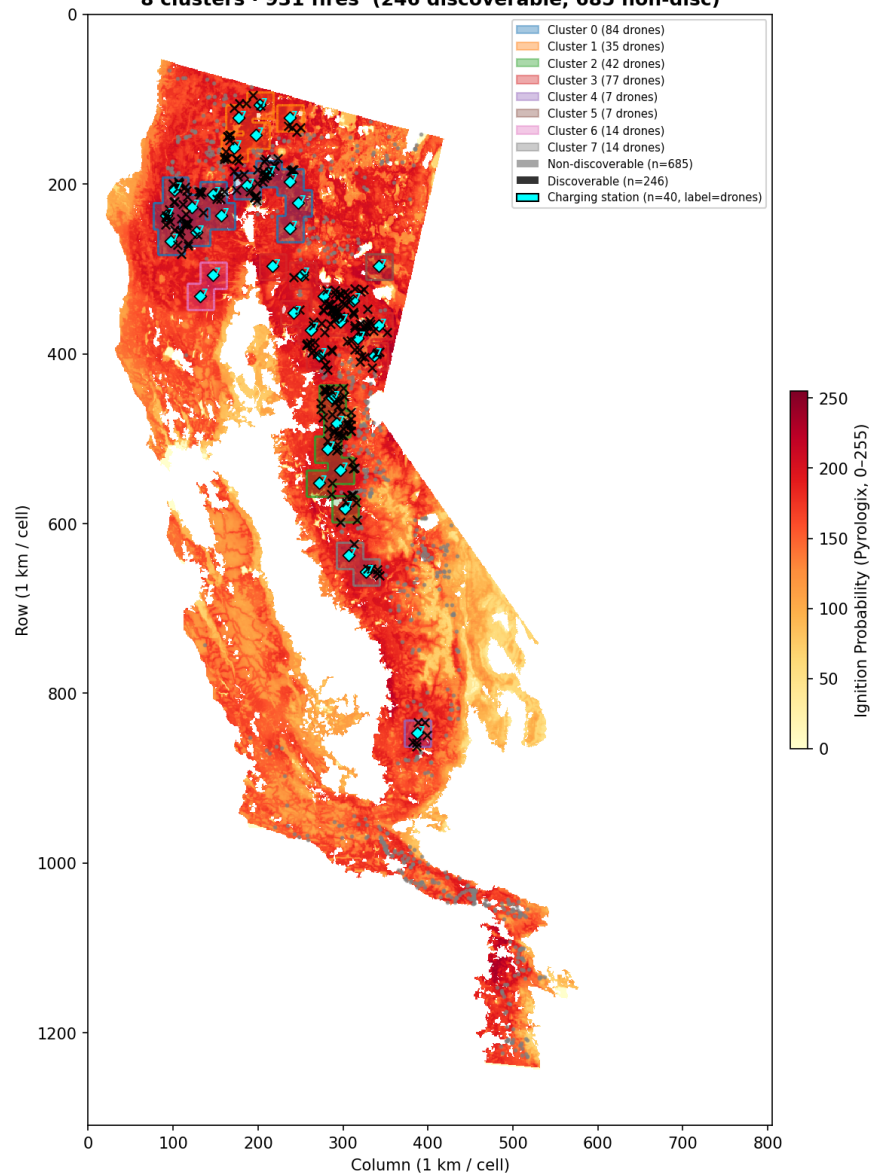


Figure 2: 20M greedy kernel — data scale

Pyrologix (opt scale) + cluster zones + fires [GaussianBudget20M_StationMax_261x161_mean]
8 clusters · 931 fires (grid 261×161, 5 km / cell)

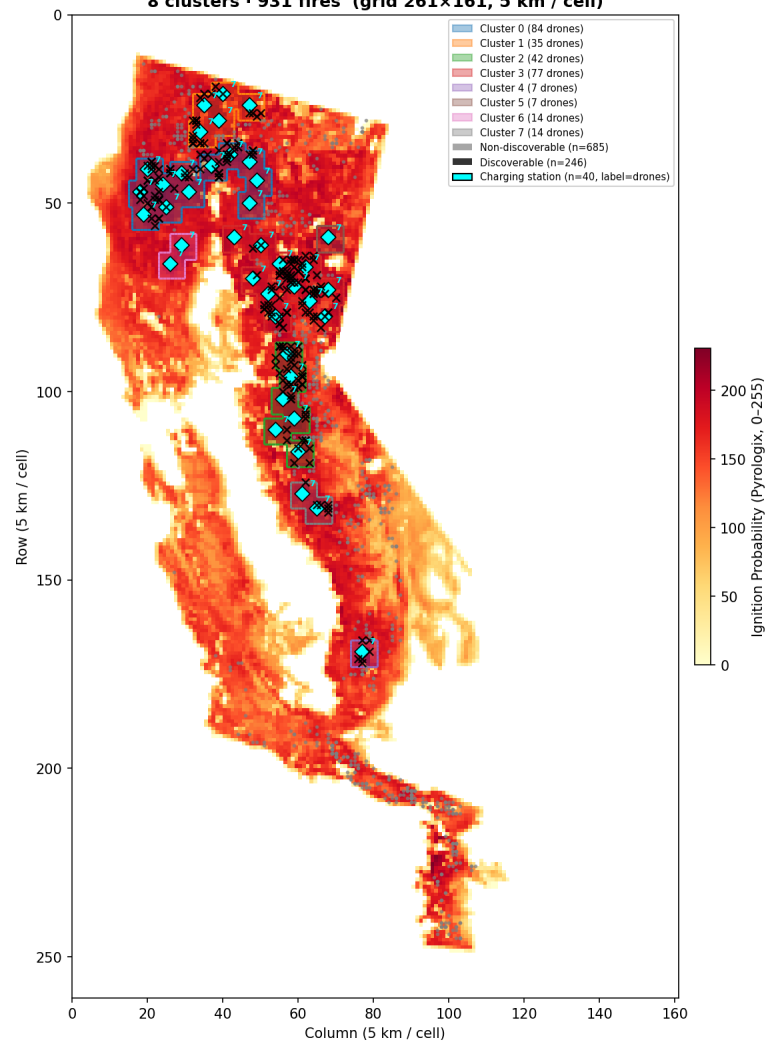


Figure 3: 20M greedy kernel — operational scale

Component	Count	Unit cost	Subtotal
Ground sensors	0	100k	0.00M
Charging stations	214	150k	32.10M
Drones	1358	50k	67.90M
Total			100.00M

The drone allocation is non-uniform, with many stations at the cap of 7 drones and a spread from 1 to 7 drones per selected station.

Solver stats

Metric	Value
Feasible station-cell pairs	122,775
Station-level drone binaries	39,158
Preprocessing	2.61 s
Model creation	6.59 s
Solving	1801.69 s
Termination	TIME_LIMIT
MIP gap	13.0%

Plots

Interpretation

At 100M, the heuristic StationMax formulation remains substantially harder than at 20M. After fixing the objective so masked cells cannot contribute and loosening the candidate pre-filter to the top **50%** of cells, the model grew substantially (122,775 feasible pairs and **39,158** station-level drone binaries) and the 30-minute run ended at **13.0%** gap. The incumbent remained a pure station+drone allocation with no ground sensors.

500M Budget

Result

The 500M run did **not** produce a meaningful placement within the 10-minute cap.

Observed outcome:

Metric	Value
Feasible station-cell pairs	239,388

Metric	Value
Station-level drone binaries	78,316
Preprocessing	3.02 s
Model creation	3.80 s
Solving	602.20 s
Termination	TIME_LIMIT
Returned incumbent	empty placement

Returned placement:

Component	Count
Ground sensors	0
Charging stations	0
Drones	0

The reported bound was positive while the incumbent objective remained at -0.0 , so the percentage gap is not meaningful. This budget should therefore be considered **unsolved** under the current 10-minute cap and formulation.

Plots

These plots are included only to show that the returned incumbent was empty.

Conclusions

1. The new greedy-kernel StationMax formulation is practical at **20M**: the model builds in a few seconds and ends with a very small gap (0.37%) under the 10-minute cap.
2. At **100M**, the same formulation becomes significantly harder than at 20M. With the fixed objective and a top-50% candidate pool, the 30-minute run ends at **13.0%** gap. This is still much better than the original 10-minute result (27.47%), but worse than the earlier 30-minute run on the smaller top-20% candidate set (9.6%).
3. At **500M**, the formulation is not usable under the 10-minute cap in its current form: no meaningful incumbent was found.
4. The preprocessing itself is **not** the bottleneck. Even with heuristic per-station kernels, preprocessing/model creation remained around 2-4 s. The difficulty comes from the MILP search at larger budgets.
5. The 100M case appears potentially usable with a larger time budget, but the candidate-set size has a strong effect on solve difficulty. The 500M case still looks too hard under the current formulation and cap.

Pyrologix + cluster zones + fires [GaussianBudget100M_StationMax_261x161_mean]
6 clusters · 931 fires (730 discoverable, 201 non-disc)

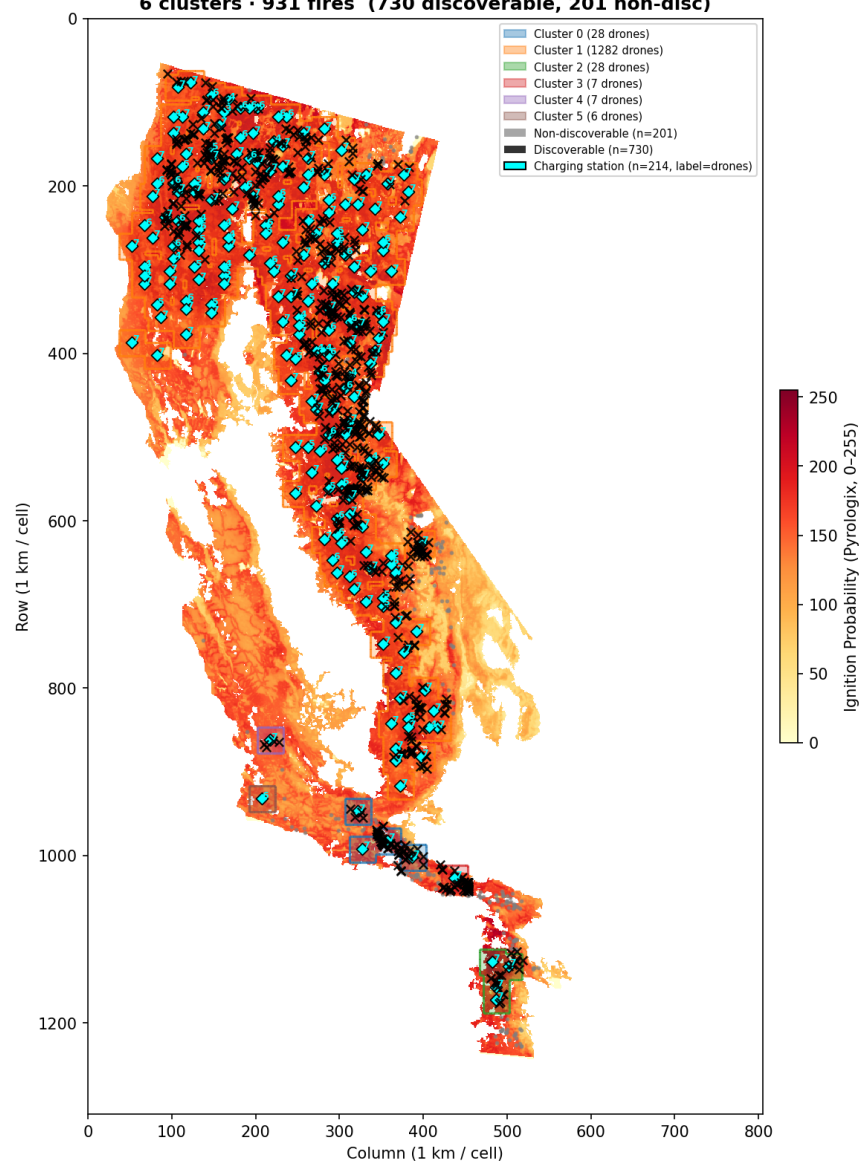


Figure 4: 100M greedy kernel — data scale

Pyrologix (opt scale) + cluster zones + fires [GaussianBudget100M_StationMax_261x161_mean]
6 clusters · 931 fires (grid 261×161, 5 km / cell)

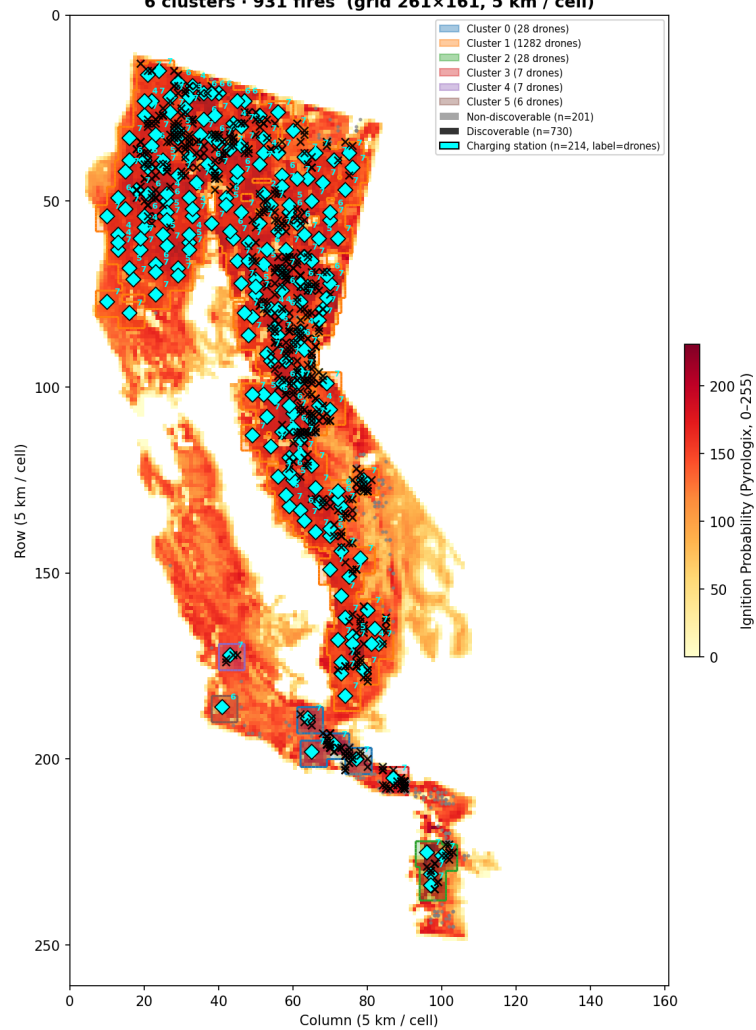


Figure 5: 100M greedy kernel — operational scale

Pyrologix + cluster zones + fires [GaussianBudget500M_StationMax_261x161_mean]
0 clusters · 931 fires (0 discoverable, 931 non-disc)

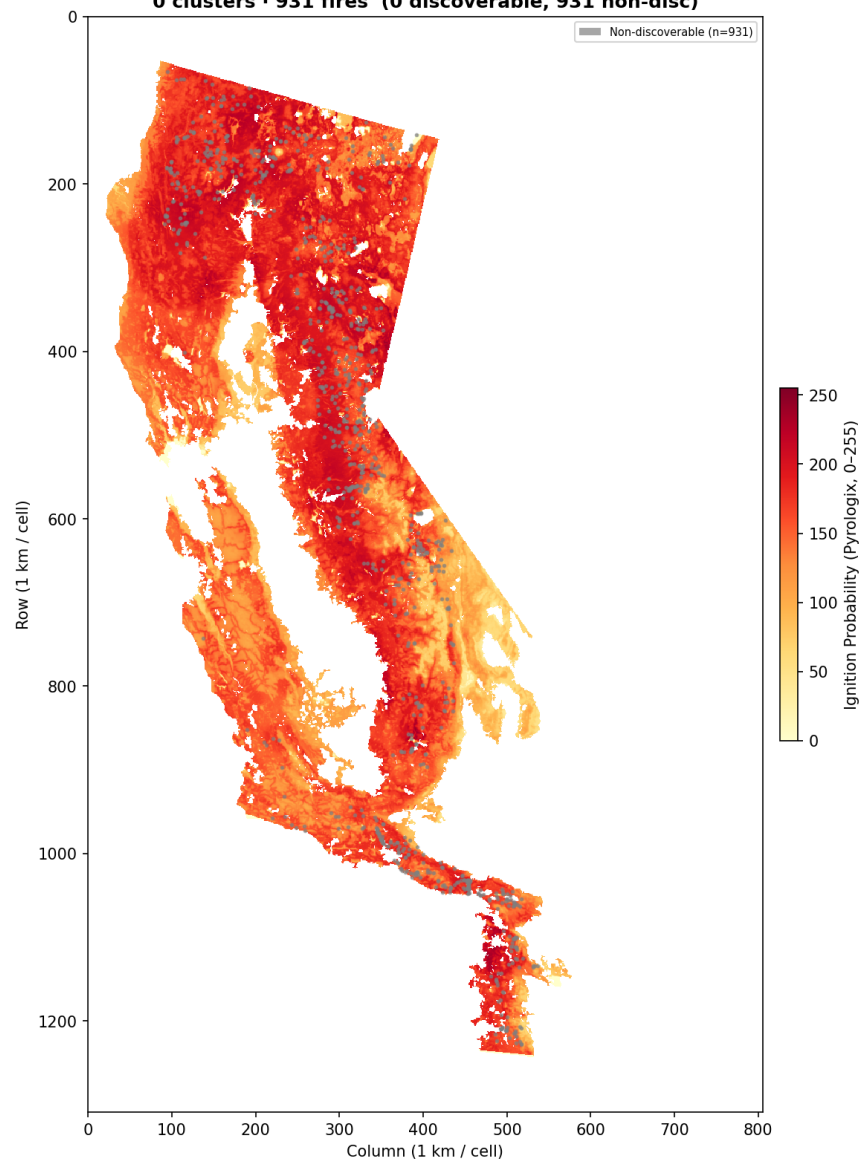


Figure 6: 500M greedy kernel — data scale

Pyrologix (opt scale) + cluster zones + fires [GaussianBudget500M_StationMax_261x161_mean]
0 clusters · 931 fires (grid 261×161, 5 km / cell)

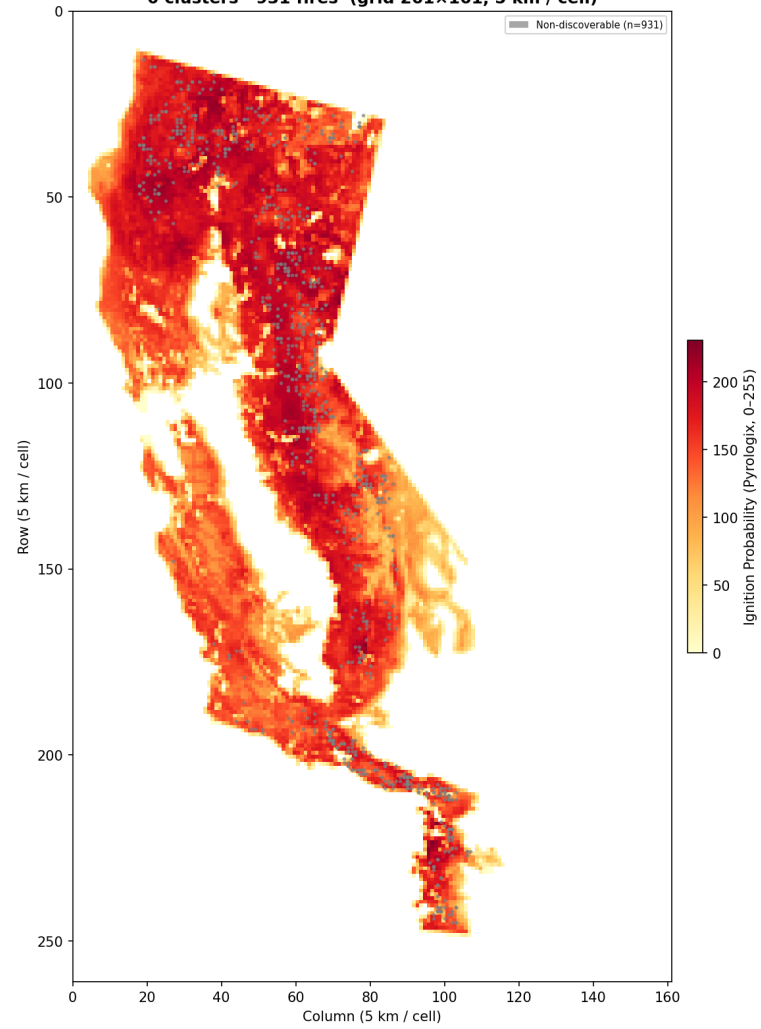


Figure 7: 500M greedy kernel — operational scale